I swear it worked on my computer! Computational Reproducibility Using Nix and rix

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2026-11-09

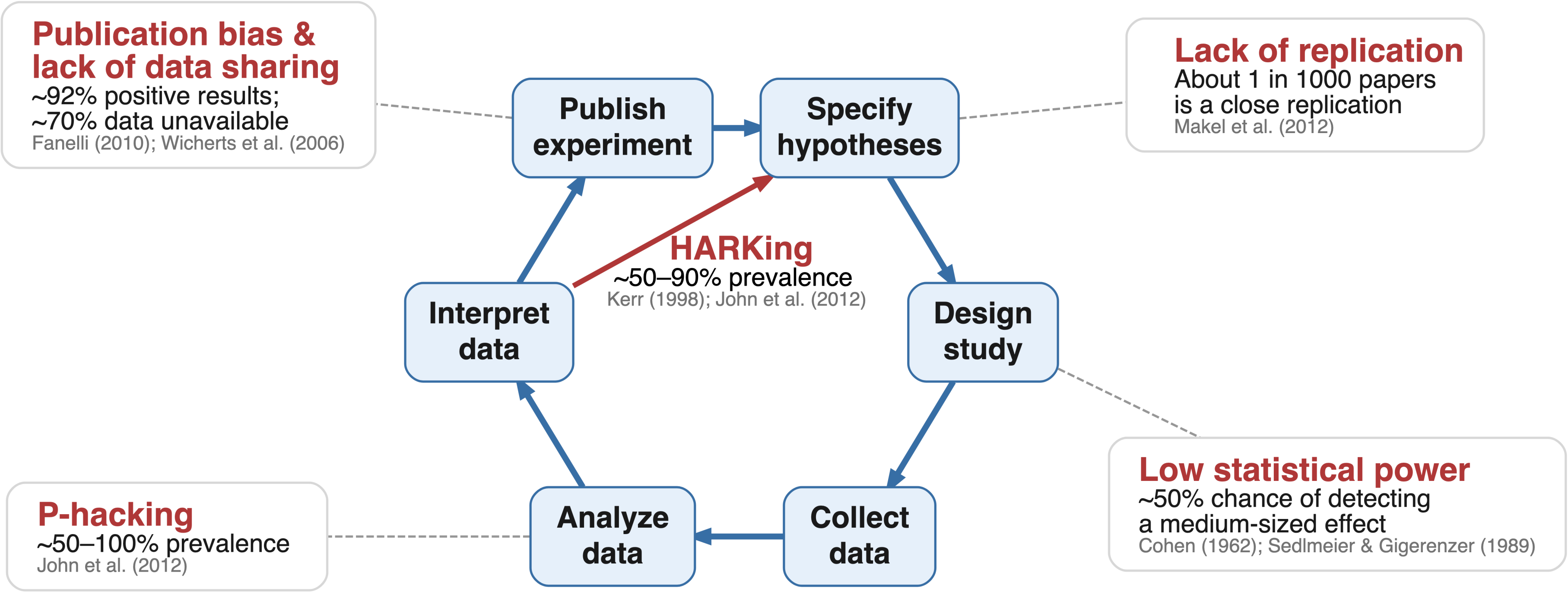
Slides

Slides: jgeller112.github.io/princeton_2026_rix-nix

Disclaimer

I will mostly be using R examples.

The Replication Crisis[1]



Publication bias & lack of data sharing
~92% positive results;
~70% data unavailable
Fanelli (2010); Wicherts et al. (2006)

Publish experiment

Specify hypotheses

Lack of replication
About 1 in 1000 papers
is a close replication
Makel et al. (2012)

HARKing
~50–90% prevalence
Kerr (1998); John et al. (2012)

Interpret data

Design study

Low statistical power
~50% chance of detecting
a medium-sized effect
Cohen (1962); Sedlmeier & Gigerenzer (1989)

P-hacking
~50–100% prevalence
John et al. (2012)

Analyze data

Collect data

~~Replication Crisis~~ Credibility Revolution!

Open Science

- Open science: being open and transparent at every stage of research[1]
- Three parts:
  - → Dissemination
  - → Stance
  - → *Workflow*: *Could someone else rerun your whole pipeline and get your results?*

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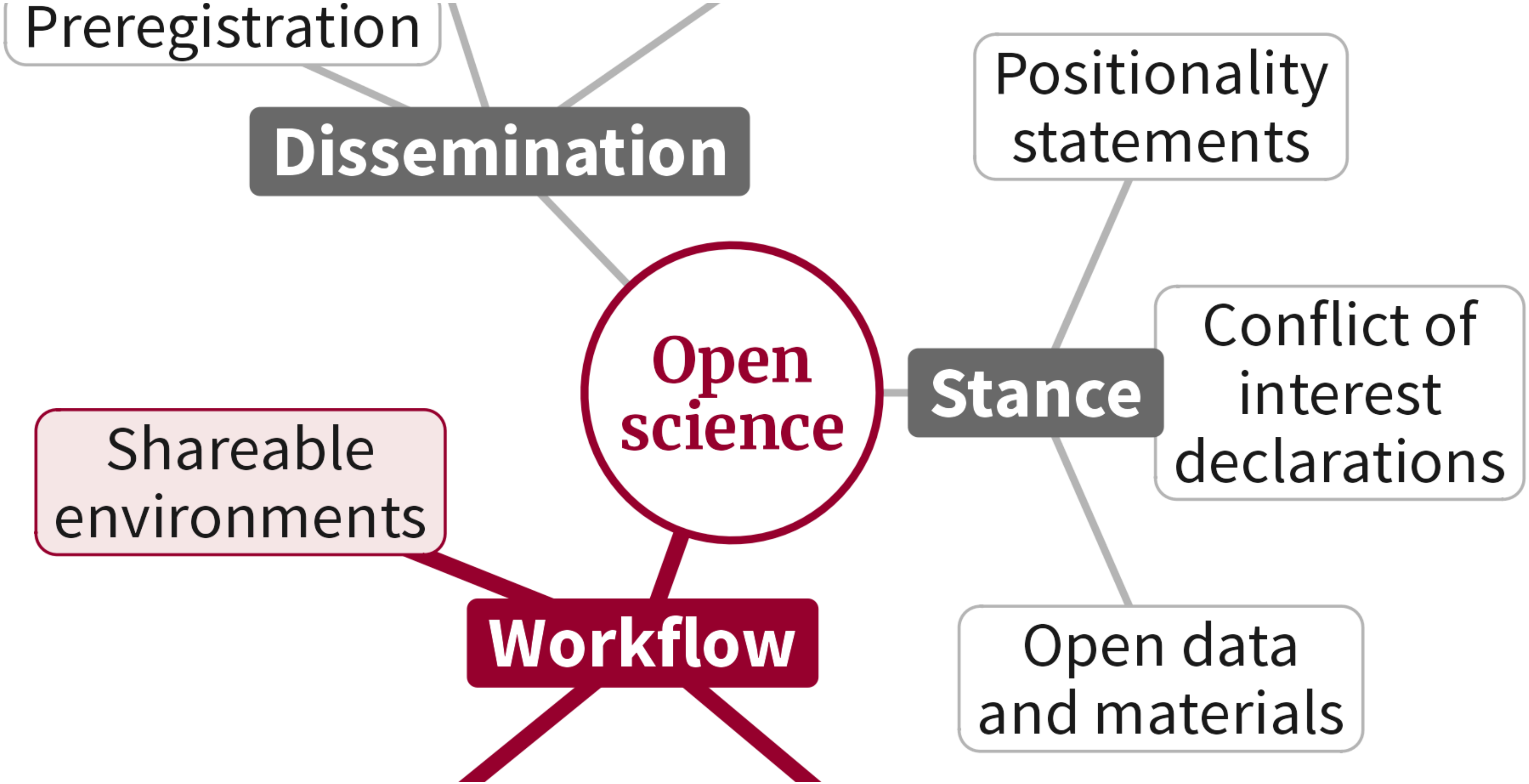
[1] Figure adapted from Casillas, J. V., Constantin-Dureci, G., Andreu Rascón, I., et al. (2025). Opening open science to all: Demystifying reproducibility and transparency practices in linguistic research. *Linguistics*, *63*(6), 1547–1575. https://doi.org/10.1515/ling-2023-0249

Open Science


Preregistration

Dissemination

Positionality statements

Open science

Stance

Conflict of interest declarations

Shareable environments

Workflow

Open data and materials

What is Reproducibility?

Analysis / Data	Same Data	Different Data
Same Analysis	Reproducible	Replicable
Different Analysis	Robust	Generalisable

Demo: everything is shared

```
study1/
├── analysis.R
└── data/
    └── data_final.csv
```

```r
# analysis.R
dat ← read.csv("data/data_final.csv")

cohens_d ← function(x) {
  m ← tapply(x$recall, x$condition, mean)
  s ← sqrt(mean(tapply(x$recall, x$condition, var)))
  unname((m["retrieval"] - m["control"]) / s)
}

set.seed(2018)
boot ← replicate(2000, cohens_d(dat[sample(nrow(dat), replace = TRUE), ]))

round(cohens_d(dat), 2)
round(quantile(boot, c(.025, .975)), 2)
```

Paper (2019): *d* = 0.45, 95% bootstrap confidence interval (CI) [0.01, 0.91], which excludes zero

Demo: same code, different year

	2019: author (R 3.5.3)	2026: me (R 4.6.1)
d	0.45	0.45
95% CI	[0.01, 0.91]	[**−0.02**, 0.88]
CI includes 0?	No	**Yes**

Why? R 3.6.0 changed how `sample()` draws, so the same seed gives different bootstrap samples.

Computational Reproducibility

> “Ability to obtain consistent computational results using the same input data, computational steps, methods, code, and conditions of analysis”[1]

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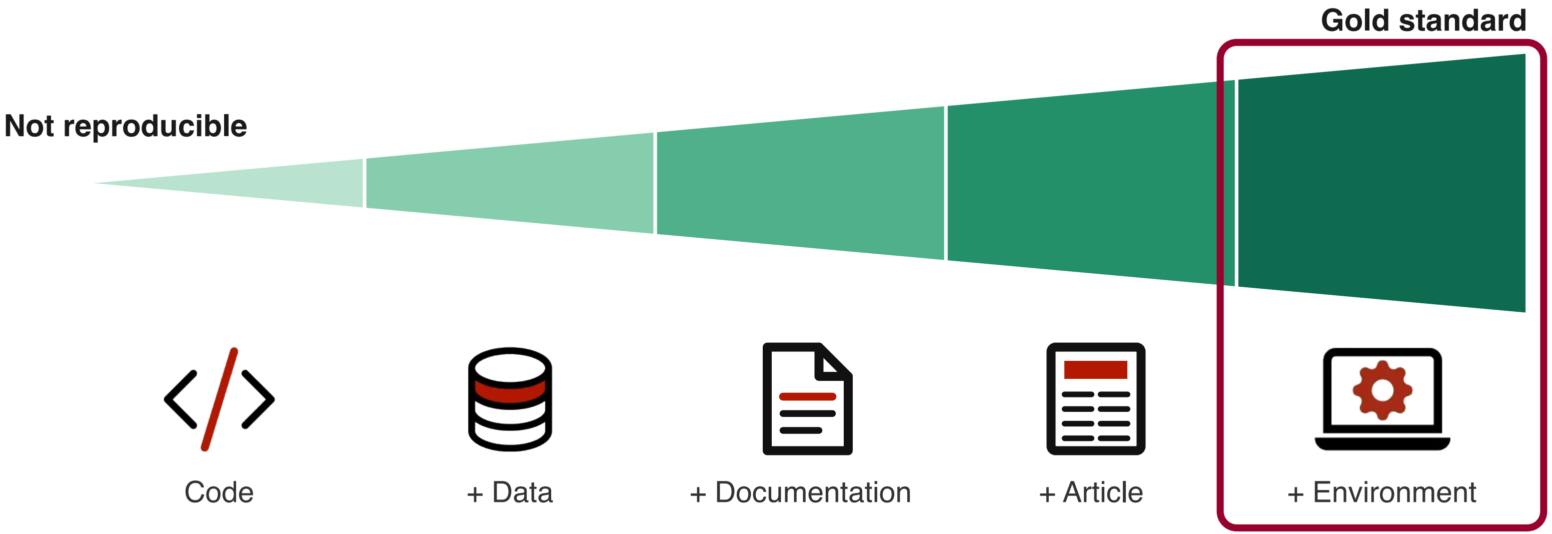
[1] Schultze, A., Saul, L., Roesch, E., et al. (2026). *Computational reproducibility: A primer from UKRN*. Zenodo. https://doi.org/10.5281/zenodo.19232891

Computational Reproducibility



Gold standard

Not reproducible

Code

+ Data

+ Documentation

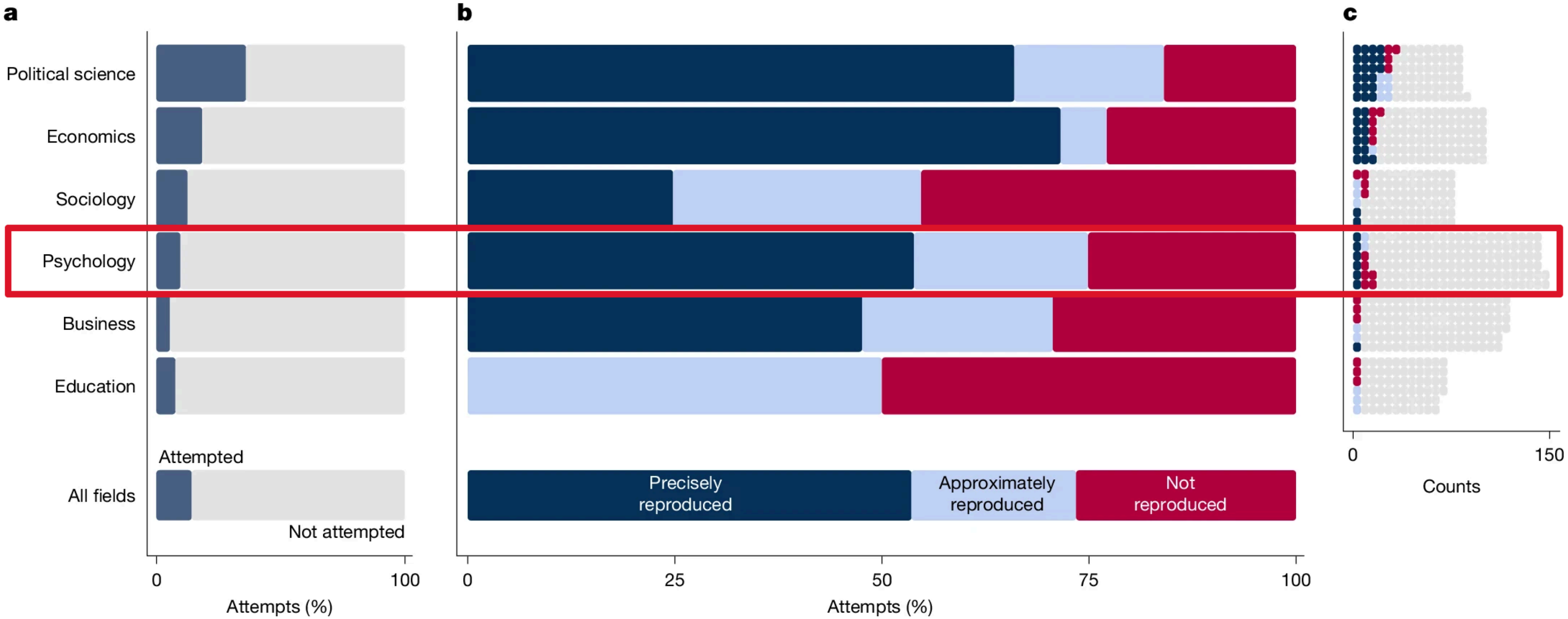
+ Article

+ Environment

Psychology: about half reproduce[1]

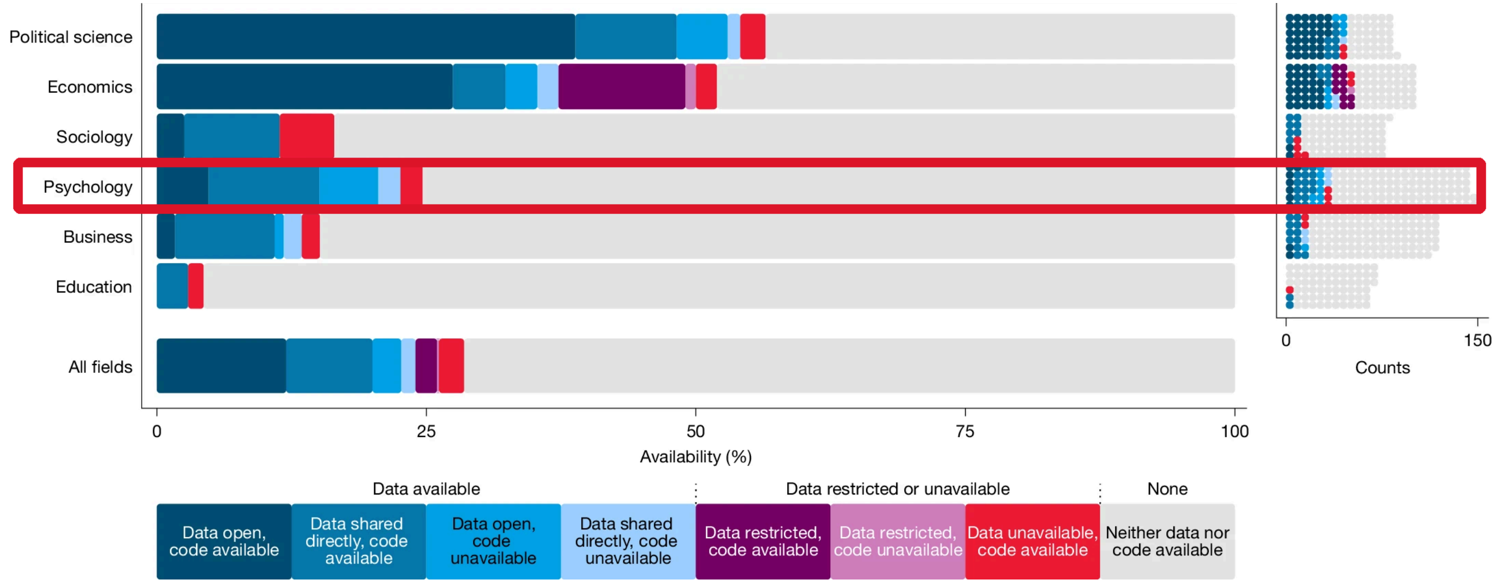
- SCORE: 600 papers, 62 journals, several fields (2009-2018)
- Out of 600 papers, 144 made data available; for 38 more, they obtained source data

Psychology: about half reproduce[1]



Why so few attempts? No data[1]

- Few psychology papers shared both data and code



Shared ≠ usable

- Quintana (2026)[1]
- Downloaded materials from OSF repos (334 papers)
  - → *Psych Sci*, *JML*, and *ESP*
  - → When materials are shared, are they usable?

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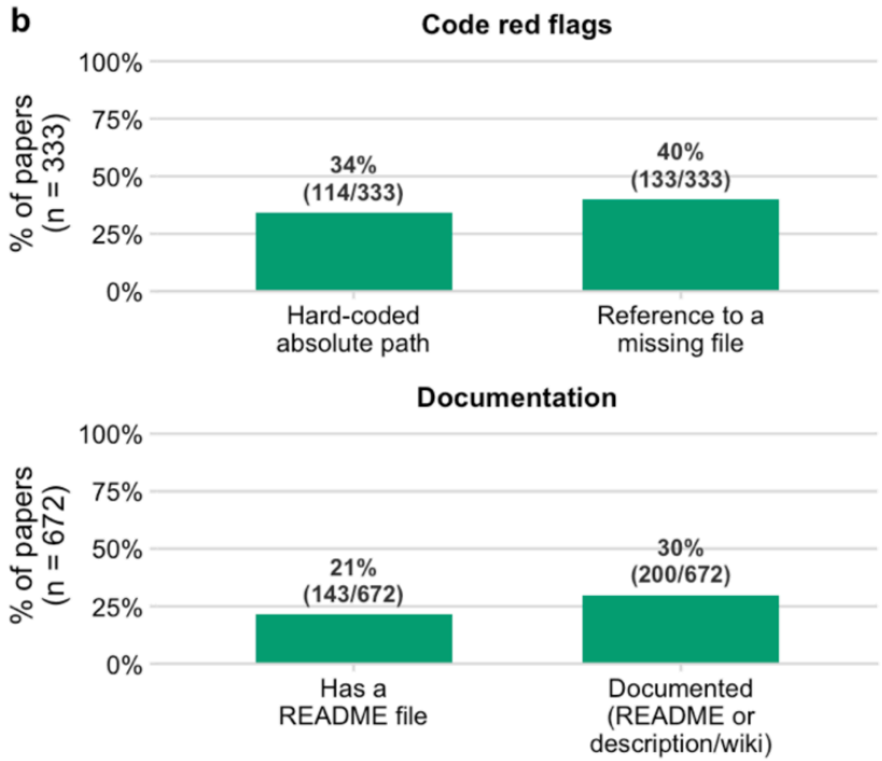
[1] Quintana, D. S. (2026, September 19). Shared research materials in psychology are widely downloaded but often difficult to reuse. PsyArXiv. https://osf.io/preprints/psyarxiv/yswku_v3

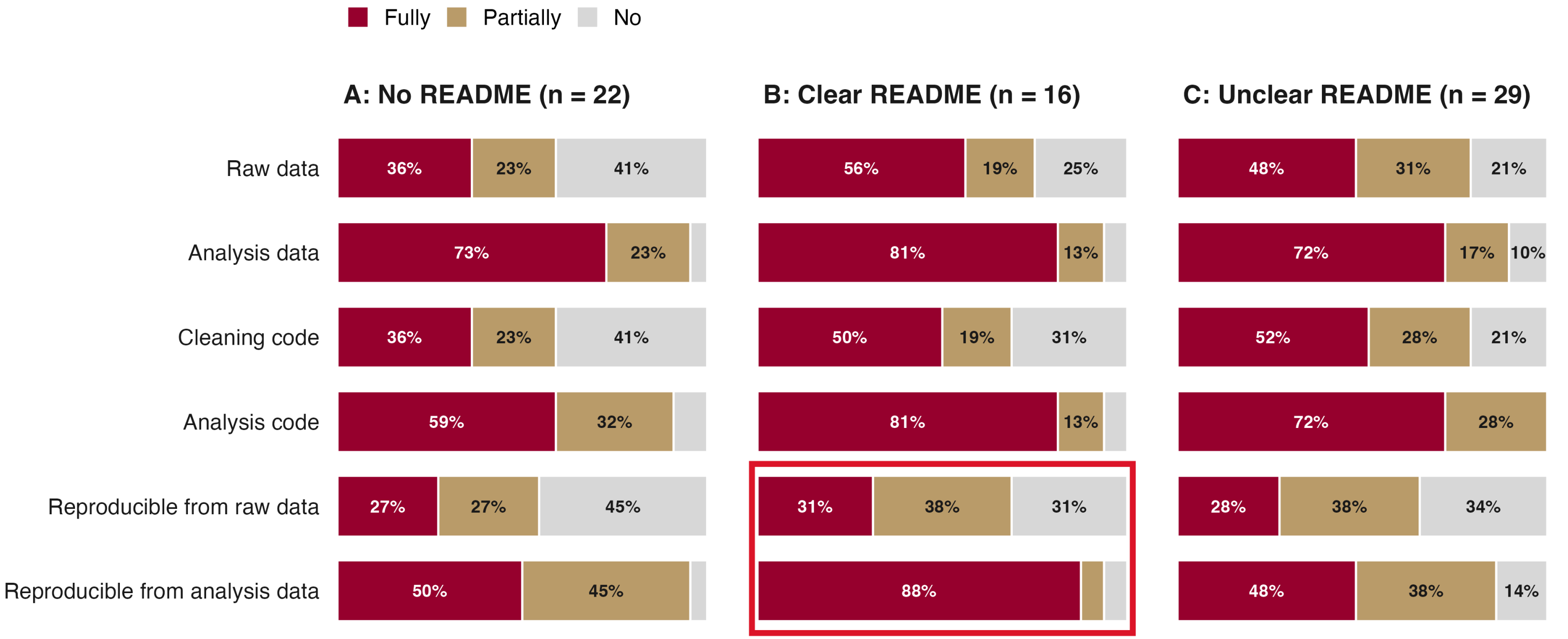
Shared ≠ usable



%

READMEs help, but only so far

- Institute for Replication[1]: 67 *Psych Science* papers (2024-2025)



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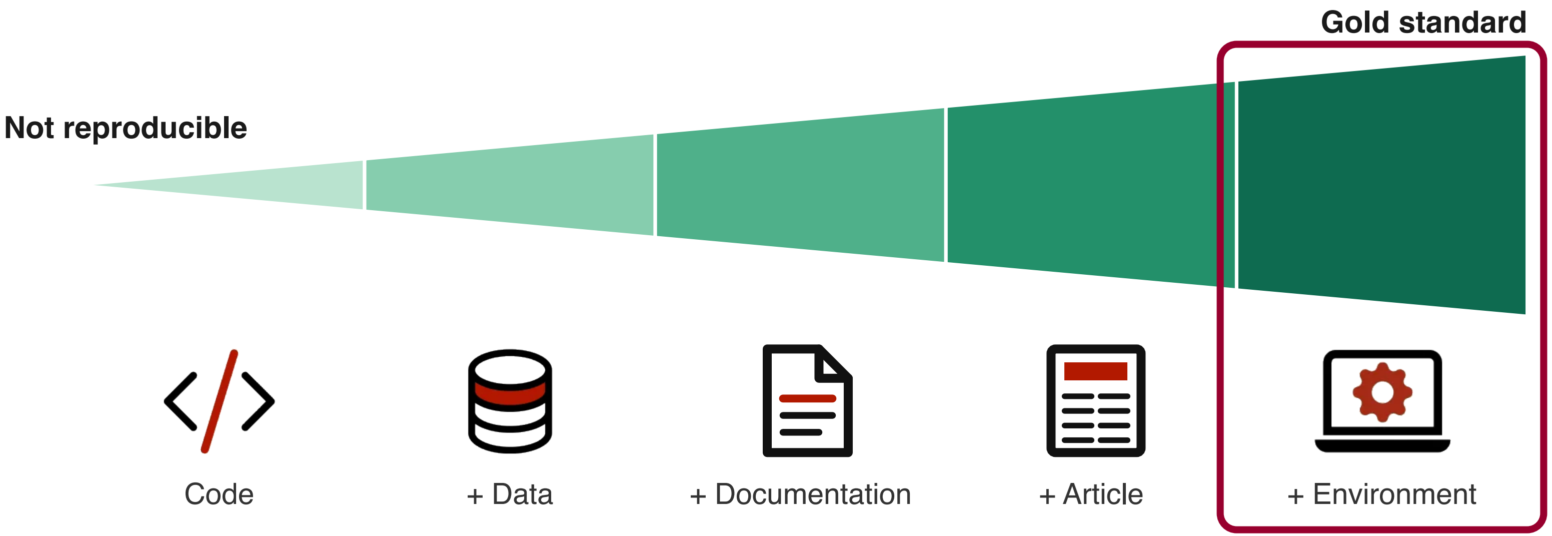
[1] Brodeur, A., Abdul Baki, G., & Fazio, L. (2026). Mass reproducibility in psychological science.

READMEs help, but only so far

- Clear README: 88% from analysis data, only 31% from raw data

Psychology: not so good 😟

- Back to the spectrum: where do these studies leave us?


Gold standard
Not reproducible
Code
+ Data
+ Documentation
+ Article
+ Environment

- Code, data, and a README get you partway. The environment is the missing step.

The Computational Environment

Which “reproducible”?[1]

Computational environment

	across space (spatial reproducibility)	**across time** (temporal reproducibility)
the environment builds (rebuildability)	builds on another machine or OS, today	the same description still builds years later
the results are the same (functional reproducibility)	same versions, same numbers, on another machine or OS	same versions, same numbers, years later
the bytes are the same (bitwise reproducibility)	identical hashes on another machine	identical hashes, years later

Environment Fragility

Setting bitwise reproducibility aside, even error-free R code sits on top of four layers, each of which may drift over space and time:

1. R version
2. R packages
3. Document tools: e.g., Pandoc, LaTeX, Quarto
4. System libraries and compilers: e.g., gcc, libpng, BLAS/LAPACK

1. R version

- Syntax: the native pipe ▷ exists only since R 4.1.0.
- Defaults: R 4.0.0 stopped turning strings into factors. Same code, different object:

```
df ← data.frame(group = c("a", "b"))
class(df$group); as.numeric(df$group)
#> R 3.6.3:  "factor"      1  2   (stringsAsFactors = TRUE)
#> R 4.0.0+: "character"  NA NA   (stringsAsFactors = FALSE)
```

- Random numbers: R 3.6.0 changed how `sample()` draws. Same seed, different draws:

```
set.seed(123); sample(1:10, 3)
#> R 3.5.3:   3  8  4    (sample.kind = "Rounding")
#> R 3.6.0+:  3 10  2    (sample.kind = "Rejection")
```

2. R packages

- Renamed or deprecated: `dplyr::cur_data()`, but now `dplyr::pick()` (warning about the future).
- Internal functions (discouraged, but used nonetheless):

3. Document tools

- Pandoc and LaTeX, called by Quarto (also outside R) or by `{rmarkdown}`. Each has its own version.
- From this talk’s demo: HTML fine, PDF failed until one more LaTeX package was declared:

```
LaTeX Error: File `lualatex-math.sty' not found.
```

4. System libraries and compilers

- Many R packages are compiled code (C, C++, Fortran) and use libraries from the OS (e.g., libpng, BLAS/LAPACK).
  - → Today: CRAN hands you a binary. However, years later: only the source. Your machine compiles it, with today's compiler and today's libraries.
- Same script, different BLAS: an eigenvector sign flip in the system's BLAS/LAPACK changes `MASS::mvrnorm()` draws (`mvtnorm::rmvnorm()` is fine). Based on an illustration by Danielle Navarro.

```
== MASS (native R) ==
  macOS:     0.28780065222640439   0.59425558497432929  -0.25961849316812602  -2.3918999759288693
  Ubuntu:   -0.43918332712576574   0.25608933342594919   0.85420518537273971  -2.2883238259260961
  Windows: -0.43918332712576580   0.25608933342594914   0.85420518537273971  -2.2883238259260965
  => DIVERGE
```

Python, too: language version

- Syntax: `match` statements exist only since Python 3.10, the walrus `:=` since 3.8.
- Random numbers: sampling from a set works in 3.10, but the draws depend on set order, which changes *every run*. In 3.11+ it errors:

```
import random; random.seed(1); random.sample({"a", "b", "c"}, 2)
#> Python 3.10:  ['a', 'b']   (or ['c', 'a'], ... depends on the run)
#> Python 3.11+: TypeError: Population must be a sequence.
```

Python, too: packages

- Changed defaults, loud: pandas 2.0 stopped silently dropping text columns:

```
df = pd.DataFrame({"group": ["a", "a", "b"],
                   "subject": ["s1", "s2", "s3"], "rt": [500, 520, 610]})
df.groupby("group").mean()
#> pandas 1.5: mean rt per group    (subject column quietly dropped)
#> pandas 2.0+: TypeError: agg function failed [how→mean,dtype→object]
```

- Changed defaults, silent: numpy 2.0 changed type promotion. No error, different number:

```
x = np.float32(0.1) + 0.2
#> numpy 1.26: 0.30000000149011613   (float64)
#> numpy 2.0+: 0.3                   (float32)
```

- Removed: `np.float` was removed in numpy 1.24, `DataFrame.append()` in pandas 2.0.
- Models: scikit-learn 0.22 changed `LogisticRegression`’s default solver (`liblinear` → `lbfgs`): same call, different coefficients.

Python, too: packages

- System libraries: numpy links OpenBLAS, MKL, or Apple Accelerate depending on how it was installed: the same BLAS story as R.

Managing the environment

Managing the environment

`{renv}` R packages

Managing the environment

Managing the environment

`{rig}` R version

uv Python + packages

conda + compiled libraries

Docker the OS

Managing the environment

Each tool pins one piece: many tools, many skill sets.

`{renv}`

- Most popular package for reproducibility, and very easy to use:
  - → `renv::init()` snapshots the packages used (and their versions) into `renv.lock`
  - → `renv::restore()` reinstalls them
- But it records, and does not restore, the version of R (can use {rig})
- But installing old packages can fail (missing OS dependencies)

{renv}

{renv} in action

```r
# analysis.R uses palmerpenguins + ggplot2
renv::init()
```

```
- Linking packages into the project
  library ... Done!
The following package(s) will be
updated in the lockfile:

# CRAN ────────────────────────────
- cli              [* → 3.6.6]
- ggplot2          [* → 4.0.3]
- palmerpenguins   [* → 0.1.1]
- rlang            [* → 1.3.0]
   ... 15 more dependencies

The version of R recorded in the lockfile
will be updated:
- R                [* → 4.6.1]

- Lockfile written to "renv.lock".
```

`renv.lock` (excerpt)

```json
{
  "R": {
    "Version": "4.6.1",
    "Repositories": [ ... ]
```

`{renv}` in action

```
  },
  "Packages": {
    "ggplot2": {
      "Package": "ggplot2",
      "Version": "4.0.3",
      "Source": "Repository",
      "Repository": "CRAN"
    },
    ...
```

R 4.6.1 is written down, not installed. `renv::restore()` uses whatever R you have.

`{renv}`: which boxes?

"use {renv} for reproducibility"

	across space (spatial reproducibility)	**across time** (temporal reproducibility)
the environment builds (rebuildability)	~ same package versions; not R or system libraries	~ old packages often fail to install
the results are the same (functional reproducibility)	~ same packages, but R and BLAS can differ	~ only if the install still works
the bytes are the same (bitwise reproducibility)	✗ not addressed	✗ not addressed

☑ covered ~ partly ☒ not covered

Docker: what a container does

Dockerfile (recipe) → image (the environment, frozen) → container (the image, running)[1]

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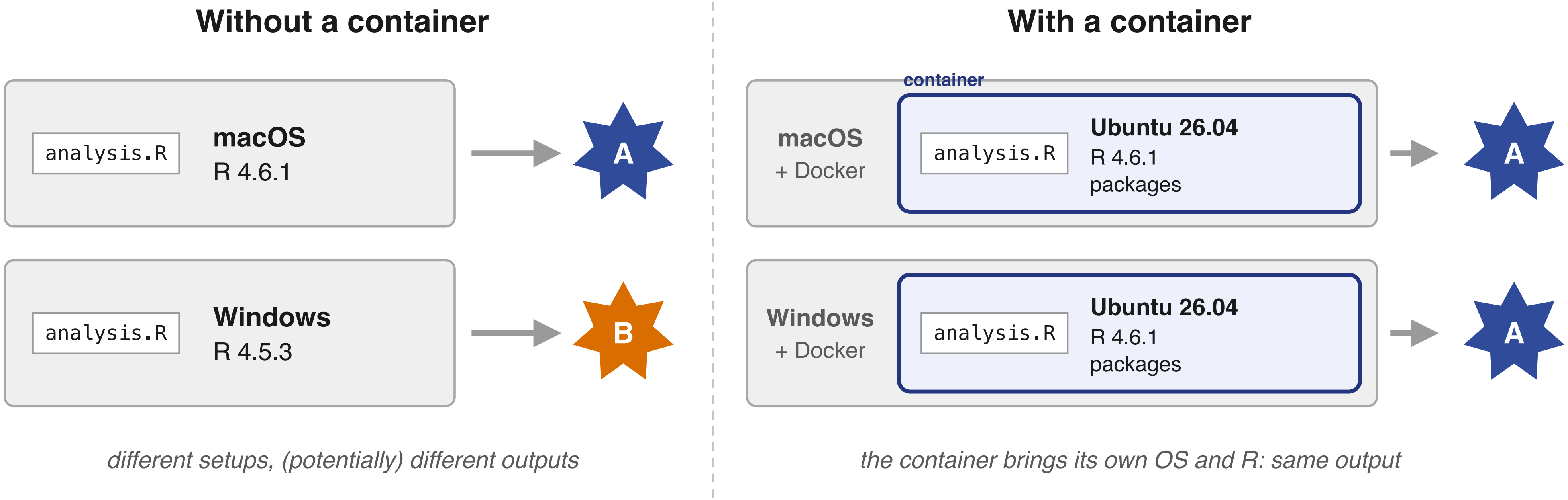
[1] Figure adapted from Rodrigues, B. *Building reproducible pipelines with R, Docker and Nix*. R-Ladies Rome (English).

Docker: what a container does



Without a container

`analysis.R` **macOS** R 4.6.1 → A

`analysis.R` **Windows** R 4.5.3 → B

different setups, (potentially) different outputs

With a container

macOS + Docker — container: `analysis.R` **Ubuntu 26.04** R 4.6.1 packages → A

Windows + Docker — container: `analysis.R` **Ubuntu 26.04** R 4.6.1 packages → A

the container brings its own OS and R: same output

Docker images

Docker images:

- contain all the software and code needed for your project
- are immutable (cannot be changed at run-time)
- can be shared on- and offline

Docker: built for shipping

- Docker’s job: make software run the same on a laptop and a server today. Reproducing a result years later was never the goal.

The recipe drifts: every line fetches what’s current *on build day*

```
FROM rocker/r-ver:latest                    # which R? whatever "latest" means today
RUN apt-get update && apt-get install -y libxml2-dev   # today's Ubuntu libraries
RUN Rscript -e 'install.packages("brms")'   # today's CRAN
```

Docker: the catches

1. Archive the image
    - Often several GB; tags on Docker Hub can move or vanish
2. Same image, different CPU
    - Runs on the host’s processor; OpenBLAS picks code per CPU, so last digits can differ
3. Costs
    - Linux know-how, slow emulation on Apple Silicon, often banned on university clusters, one company in charge

Docker: which boxes?

"Docker guarantees full computational reproducibility"

	across space (spatial reproducibility)	**across time** (temporal reproducibility)
the environment builds (rebuildability)	✓ pull the saved image anywhere	~ drifts unless you pin the base image + use renv or Nix inside
the results are the same (functional reproducibility)	✓ same image, same versions	~ only if you archived the image
the bytes are the same (bitwise reproducibility)	~ CPU-dependent BLAS can differ	✗ not addressed

✓ covered ~ partly ✗ not covered

Nix/{Rix}

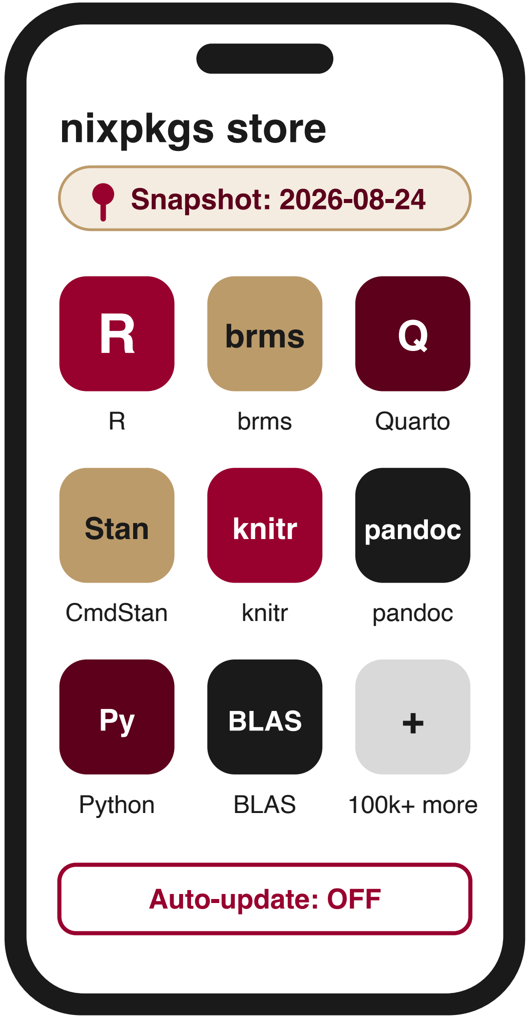
- Nix is a package manager (+language +build tool):
  - it installs and manages R/Python packages and software/system dependencies
  - Built for reproducible builds: R, R packages *and* system dependencies, all declared in one text file (a *Nix expression*)
  - Every input is pinned by hash: the same expression gives the same environment, on any machine (everything lives in its own folder)
    - The “+ Environment” step of the gold standard

Logo: NixOS artwork (CC-BY 4.0)

Nix is like an app store

- Like an app store on your phone, but for research software
- One catalog (`nixpkgs`): R, R packages, Quarto, CmdStan, system libraries
- Your `default.nix` is the list of “apps” to install
- The twist: the store is frozen on a date. No auto-updates, so everyone gets the exact same versions

Nix is like an app store


nixpkgs store
R
R
brms
brms
Q
Quarto
Stan
CmdStan
knitr
knitr
pandoc
pandoc
Py
Python
BLAS
BLAS
+
100k+ more

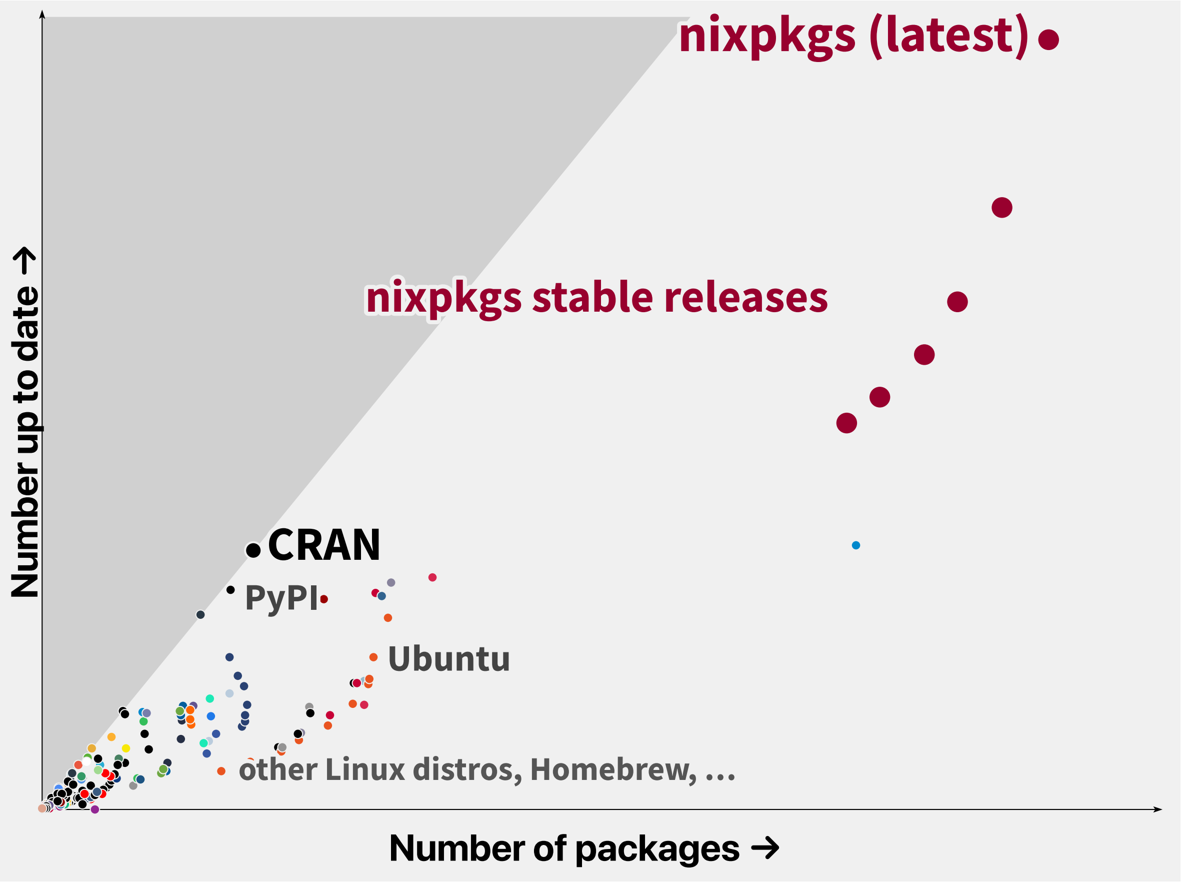
Snapshot: 2026-08-24

Auto-update: OFF

How big is `nixpkgs`?[1]

- `nixpkgs` (top right): biggest, freshest; nearly all of CRAN (search)

How big is `nixpkgs`?[1]



Nix: the empirical evidence

Malka, Zacchiroli, & Zimmermann (2025)[1] rebuilt 709,816 packages from `nixpkgs` snapshots taken every four months, 2017–2023:

- Rebuildable: 99.7% to 99.9% still build from their description.
- Bitwise reproducible: 69% in 2017, 91% in 2023, rising.
- Functionally reproducible: not measured, but follows for every package that rebuilds. From my understanding, every input is pinned by hash, so a rebuild can fail but cannot succeed with different versions. This is not the same with Docker.

---

[1] Malka, J., Zacchiroli, S., & Zimmermann, T. (2025). Does functional package management enable reproducible builds at scale? Yes. *2025 IEEE/ACM 22nd International Conference on Mining Software Repositories (MSR)*. https://doi.org/10.1109/MSR66628.2025.00115

Nix: which boxes?

“Nix: every input pinned by hash”

	across space (spatial reproducibility)	**across time** (temporal reproducibility)
the environment builds (rebuildability)	✓ same default.nix, any machine	✓ 99.7–99.9% still rebuild
the results are the same (functional reproducibility)	✓ same pinned versions everywhere	✓ same pinned inputs, years later
the bytes are the same (bitwise reproducibility)	~ mostly: 91% bitwise in 2023	~ rising: 69% (2017) → 91% (2023)

✓ covered ~ partly ✗ not covered

Nix: a Nix expression (`default.nix`)

```nix
let
  pkgs = import (fetchTarball
    "https://github.com/rstats-on-nix/nixpkgs/archive/2026-08-24.tar.gz") {};
  rpkgs = builtins.attrValues {
    inherit (pkgs.rPackages) brms knitr palmerpenguins rmarkdown;
  };
  cmdstanr = pkgs.rPackages.buildRPackage {
    name = "cmdstanr";
    src = pkgs.fetchgit {
      url    = "https://github.com/stan-dev/cmdstanr";
      rev    = "da99e2ba954658bdad63bffb738c4444c33a4e0e";
      sha256 = "sha256-wXfOxBexnuL83fvCM+6qv6d7UhTLGq+Xhvja3lRfQpI=";
    };
  };
  system_packages = builtins.attrValues {
    inherit (pkgs) cmdstan pandoc quarto R;
  };
  shell = pkgs.mkShell {
    buildInputs = [ cmdstanr rpkgs system_packages ];
    shellHook   = "export CMDSTAN=...";
  };
in { inherit pkgs shell; }
```

The snapshot: every package definition comes from nixpkgs as it was on 2026-08-24.

Nix: a Nix expression (`default.nix`)

```nix
let
  pkgs = import (fetchTarball
    "https://github.com/rstats-on-nix/nixpkgs/archive/2026-08-24.tar.gz") {};
  rpkgs = builtins.attrValues {
    inherit (pkgs.rPackages) brms knitr palmerpenguins rmarkdown;
  };
  cmdstanr = pkgs.rPackages.buildRPackage {
    name = "cmdstanr";
    src = pkgs.fetchgit {
      url    = "https://github.com/stan-dev/cmdstanr";
      rev    = "da99e2ba954658bdad63bffb738c4444c33a4e0e";
      sha256 = "sha256-wXfOxBexnuL83fvCM+6qv6d7UhTLGq+Xhvja3lRfQpI=";
    };
  };
  system_packages = builtins.attrValues {
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  };
  shell = pkgs.mkShell {
    buildInputs = [ cmdstanr rpkgs system_packages ];
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```

R packages from that snapshot: exact versions, plus all of their dependencies.

Nix: a Nix expression (`default.nix`)

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  };
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    name = "cmdstanr";
    src = pkgs.fetchgit {
      url    = "https://github.com/stan-dev/cmdstanr";
      rev    = "da99e2ba954658bdad63bffb738c4444c33a4e0e";
      sha256 = "sha256-wXfOxBexnuL83fvCM+6qv6d7UhTLGq+Xhvja3lRfQpI=";
    };
  };
  system_packages = builtins.attrValues {
    inherit (pkgs) cmdstan pandoc quarto R;
  };
  shell = pkgs.mkShell {
    buildInputs = [ cmdstanr rpkgs system_packages ];
    shellHook   = "export CMDSTAN=...";
  };
in { inherit pkgs shell; }
```

cmdstanr from GitHub: exact commit *and* content hash. If GitHub changes, the build fails rather than silently differing. By hand, you compute that hash yourself.

Nix: a Nix expression (`default.nix`)

```nix
let
  pkgs = import (fetchTarball
    "https://github.com/rstats-on-nix/nixpkgs/archive/2026-08-24.tar.gz") {};
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    inherit (pkgs.rPackages) brms knitr palmerpenguins rmarkdown;
  };
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    src = pkgs.fetchgit {
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      rev    = "da99e2ba954658bdad63bffb738c4444c33a4e0e";
      sha256 = "sha256-wXfOxBexnuL83fvCM+6qv6d7UhTLGq+Xhvja3lRfQpI=";
    };
  };
  system_packages = builtins.attrValues {
    inherit (pkgs) cmdstan pandoc quarto R;
  };
  shell = pkgs.mkShell {
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    shellHook   = "export CMDSTAN=…";
  };
in { inherit pkgs shell; }
```

System software: R itself, Quarto, pandoc and CmdStan, from the same snapshot.

Nix: a Nix expression (`default.nix`)

```nix
let
  pkgs = import (fetchTarball
    "https://github.com/rstats-on-nix/nixpkgs/archive/2026-08-24.tar.gz") {};
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    };
  };
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  shell = pkgs.mkShell {
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    shellHook   = "export CMDSTAN=…";
  };
in { inherit pkgs shell; }
```

The shell: everything above, bundled into what `nix-shell` opens. By hand, this means learning Nix. `{rix}` writes it for you, from R.

Nix/`Rix`

- `{rix}`, by Bruno Rodrigues (website), makes writing Nix expressions easy: one R function, `rix()`

```r
install.packages("rix", repos = "https://ropensci.r-universe.dev")
```

Nix/`Rix`: the `rix()` call

```r
library(rix)

rix(
  date    = "2026-08-24",
  r_pkgs = c("brms", "palmerpenguins", "knitr", "rmarkdown"),
  git_pkgs = list(
    package_name = "cmdstanr",
    repo_url     = "https://github.com/stan-dev/cmdstanr",
    commit       = "da99e2ba954658bdad63bffb738c4444c33a4e0e"
  ),
  system_pkgs = c("quarto", "cmdstan"),
  shell_hook  = paste(
    "export CMDSTAN=$(dirname $(dirname $(command -v stan)))/opt/cmdstan",
    "export PRECOMPILED_HEADERS=false",
    sep = "\n"
  ),
  ide          = "none",
  project_path = ".",
  overwrite    = TRUE
)
```

`date`: becomes the snapshot line in `default.nix`, pinning *everything* to one snapshot of nixpkgs. R and every package come in the versions they had on that day, on any machine, forever.

Nix/`Rix`: the `rix()` call

```r
library(rix)

rix(
  date    = "2026-08-24",
  r_pkgs = c("brms", "palmerpenguins", "knitr", "rmarkdown"),
  git_pkgs = list(
    package_name = "cmdstanr",
    repo_url     = "https://github.com/stan-dev/cmdstanr",
    commit       = "da99e2ba954658bdad63bffb738c4444c33a4e0e"
  ),
  system_pkgs = c("quarto", "cmdstan"),
  shell_hook  = paste(
    "export CMDSTAN=$(dirname $(dirname $(command -v stan)))/opt/cmdstan",
    "export PRECOMPILED_HEADERS=false",
    sep = "\n"
  ),
  ide          = "none",
  project_path = ".",
  overwrite    = TRUE
)
```

`r_pkgs`: CRAN packages, at their versions on that date. `knitr` and `rmarkdown` are here because Quarto needs them to run R code.

Nix/`Rix`: the `rix()` call

```r
library(rix)

rix(
  date    = "2026-08-24",
  r_pkgs = c("brms", "palmerpenguins", "knitr", "rmarkdown"),
  git_pkgs = list(
    package_name = "cmdstanr",
    repo_url     = "https://github.com/stan-dev/cmdstanr",
    commit       = "da99e2ba954658bdad63bffb738c4444c33a4e0e"
  ),
  system_pkgs = c("quarto", "cmdstan"),
  shell_hook  = paste(
    "export CMDSTAN=$(dirname $(dirname $(command -v stan)))/opt/cmdstan",
    "export PRECOMPILED_HEADERS=false",
    sep = "\n"
  ),
  ide          = "none",
  project_path = ".",
  overwrite    = TRUE
)
```

`git_pkgs`: packages that aren’t on CRAN. cmdstanr is pinned to an exact commit (the v0.9.0 release), not “whatever is on GitHub today”.

Nix/`Rix`: the `rix()` call

```r
library(rix)

rix(
  date    = "2026-08-24",
  r_pkgs = c("brms", "palmerpenguins", "knitr", "rmarkdown"),
  git_pkgs = list(
    package_name = "cmdstanr",
    repo_url     = "https://github.com/stan-dev/cmdstanr",
    commit       = "da99e2ba954658bdad63bffb738c4444c33a4e0e"
  ),
  system_pkgs = c("quarto", "cmdstan"),
  shell_hook  = paste(
    "export CMDSTAN=$(dirname $(dirname $(command -v stan)))/opt/cmdstan",
    "export PRECOMPILED_HEADERS=false",
    sep = "\n"
  ),
  ide          = "none",
  project_path = ".",
  overwrite    = TRUE
)
```

`system_pkgs`: software outside R. Quarto and CmdStan are the two things that broke in the demo; now they’re part of the environment.

Nix/`Rix`: the `rix()` call

```r
library(rix)

rix(
  date    = "2026-08-24",
  r_pkgs = c("brms", "palmerpenguins", "knitr", "rmarkdown"),
  git_pkgs = list(
    package_name = "cmdstanr",
    repo_url     = "https://github.com/stan-dev/cmdstanr",
    commit       = "da99e2ba954658bdad63bffb738c4444c33a4e0e"
  ),
  system_pkgs = c("quarto", "cmdstan"),
  shell_hook  = paste(
    "export CMDSTAN=$(dirname $(dirname $(command -v stan)))/opt/cmdstan",
    "export PRECOMPILED_HEADERS=false",
    sep = "\n"
  ),
  ide          = "none",
  project_path = ".",
  overwrite    = TRUE
)
```

`shell_hook`: commands that run each time you enter the environment. Here: tell cmdstanr where CmdStan lives.

Nix/`Rix`: the `rix()` call

```r
library(rix)

rix(
  date   = "2026-08-24",
  r_pkgs = c("brms", "palmerpenguins", "knitr", "rmarkdown"),
  git_pkgs = list(
    package_name = "cmdstanr",
    repo_url     = "https://github.com/stan-dev/cmdstanr",
    commit       = "da99e2ba954658bdad63bffb738c4444c33a4e0e"
  ),
  system_pkgs = c("quarto", "cmdstan"),
  shell_hook  = paste(
    "export CMDSTAN=$(dirname $(dirname $(command -v stan)))/opt/cmdstan",
    "export PRECOMPILED_HEADERS=false",
    sep = "\n"
  ),
  ide          = "none",
  project_path = ".",
  overwrite    = TRUE
)
```

`ide`, `project_path`, `overwrite`: no IDE (`\"rstudio\"`, `\"positron\"`, `\"code\"` also work); write `default.nix` into this folder.

Nix/`Rix`: Python too

```r
rix(
  date     = "2026-08-24",
  r_pkgs   = c("brms", "reticulate"),
  py_conf = list(
    py_version = "3.12",
    py_pkgs    = c("pandas", "numpy")
  ),
  ide          = "none",
  project_path = ".",
  overwrite    = TRUE
)
```

```
$ nix-shell --pure --run 'python --version'
Python 3.12.14                  # pandas 3.0.4, numpy 2.5.1: same 2026-08-24 snapshot
```

`py_conf`: one `default.nix` pins R, Python *and* both sets of packages. R reaches that exact Python through `reticulate`, never a system one.

Nix/`Rix`: using it

Install Nix (macOS, Linux, or Windows via WSL2), using the Determinate Systems installer:

```
curl --proto '=https' --tlsv1.2 -sSf -L https://install.determinate.systems/nix | \
  sh -s -- install
```

```
Rscript gen-env.R    # rix() writes default.nix
nix-build            # build it (slow only the first time)
nix-shell --pure --run "quarto render nix_rix_example.qmd"
```

- Plain `nix-shell` drops you into the environment to work interactively
- `--pure`: nothing from your own machine leaks in: not your R, not your packages, not `~/.cmdstan`

Illustration

- https://github.com/jgeller112/Rix-Nix_Test

Nix/`Rix`: using it (recording)

Nix not perfect

- Misdeclared dependencies
  - → A package lists a dependency under Imports that belongs in Suggests
- macOS
  - → The recipe stays valid, but an OS update can break today's build
  - → Recommendation: build inside Docker
- Isolation
  - → Use `nix-shell --pure`: only the declared environment, nothing from the host
- Windows
  - → {rix} can be cumbersome on Windows: you need WSL

What a Lab Can Actually Do

Not ready for Nix/{Rix}? Start smaller:

- {renv} and {rig} for R dependency snapshots and R version management
  - → {groundhog}
- Docker for a shared environment
- {targets} for analytic pipelines
- Metacheck (https://www.scienceverse.org/metacheck_book/) allows for a reproduibility check

Culture, Not Just Tools

- Train RAs
- Make “can a stranger rerun this?” a lab norm
- Include tools/discussion in courses

Takeaways

1. Reproducibility is a ladder, not a cliff
    - Every step you take makes your pipeline more reproducible
2. Document it
    - A clear README: what to run, in what order, starting from raw data
3. Freeze the environment
    - Nix/`{rix}` declares R, packages, and system tools in one file, pinned to a date

Thanks!

- Interested in learning more about Nix and {rix} check out my tutorial paper with Felipe Vieira and Bruno Rodrigues:

i Forthcoming in *AMPPS*

Vieira, F. F., Geller, J., & Rodrigues, B. (2026, January 28). Why risk it, when you can {rix} it: A tutorial for computational reproducibility focused on simulation studies. Retrieved from osf.io/preprints/psyarxiv/a3e8v_v1.